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LM-79-08 Test Report

For

Torshare Ltd.

(Brand Name: TORSHARE)

5th Minying Road No.3 Industrial Area Shuitian Shiyan Town Bao'an District, Shenzhen, Guangdong, 518108

Model name(s): TS-ST4-8FT80W-H1-TCP[Blank;(EM),(EMS),(MS)]

Report Type:	Testing and Report According to IES LM-79-2008
Type of Luminaire:	Direct Linear Ambient Luminaires
Report Date:	2023-08-24
Prepared By:	Ningbo TengLi Testing Co., Ltd 2nd floor, Block B, Ningbo Testing and Certification Base, No. 66 Qingyi Road, Ningbo National Hi-Tech Zone, Ningbo, Zhejiang

Test & Report By:

Engineer: Wat Wang

Review By:

Manager: Nick Song

Note: 1. The results contained in this report pertain only to the tested samples

2. This report does not imply product certification, approval, or endorsement by A2LA, or any agency of the Federal Government.

1.1 Product Information:		
Model Number	TS-ST4-8FT80W-H1-TCP[Blank;(EM),(EMS),(MS)]	
Remark	The [Blank;(MS),(EM),(EMS)] can be MS=with Motion Sensor, EM=with Emergency , EMS=with Motion Sensor and Emergency, Blank=without Motion Sensor and Emergency.	
Representative (Tested) Model	TS-ST4-8FT80W-H1-TCP(0%,3500K) TS-ST4-8FT80W-H1-TCP(50%,5000K) TS-ST4-8FT80W-H1-TCP(100%,6500K)	
Note	The construction of Model TS-ST4-4FT40W-H1-TCP is similar with TS-ST4-8FT80W-H1-TCP, except for length and wattage. The IES of TS-ST4-8FT80W-H1-TCP was calculated from the IES of TS-ST4-4FT40W-H1-TCP. The data of Model TS-ST4-4FT40W-H1-TCP comes from report JDE230802-A	
Model Difference	N/A	
SKU (if available)	N/A	
Type of Luminaire (for integral lamps, list base type and lamp type)	Direct Linear Ambient Luminaires	
LED Manufacturer	Bridgelux,Inc.	
LED Model	BXEN-35E-21M-3C BXEN-65E-21M-3C	
Dimming	Continuous	
Integral Controls	Yes	
Sample Number	JDE230802-B1	
Date of Receipt	Aug.14,2023	
Luminaire Aperture (for downlights)	--	in.
Luminaire Length	--	mm
Luminaires Width	--	mm
Number of Units (modular products)	N/A	s

1.2 Rated Values:	
Rated Voltage / Frequency	120-277Vac, 50/60Hz
Nominal Power	50W/64W/80W (Power Adjustable)
Rated Initial Lamp Lumen	--
Declared CCT	3500K/4000K/5000K/5700K/6500K (Color Tunable)

1.3 Test Specifications:

Test item	1. Total Luminous Flux 2. Luminous Distribution Intensity 3. Luminous Efficacy 4. Correlated Color Temperature 5. Color Rendering Index 6. Chromaticity Coordinate 7. Electrical Parameters
Reference Standard	1. IES LM-79-2008 Electrical and Photometric Measurements of Solid-State Lighting Products 2. ANSI C78.377-2008 Specifications for the Chromaticity of Solid State Lighting Products 3. CIE 13.3-1995 Method of Measuring and Specifying Colour Rendering Properties of Light Sources 4. CIE 15-2004 Technical Report Colorimetry 5. IESNA LM-16-93 Practical Guide to Colorimetry of Light Source

1.4 Test Methods

1) Photometric and Light Distribution Measurement – Goniophotometer Method: Photometric parameters were measured using the goniophotometer and software. The ambient temperature shall be maintained at 25 ℃ ±1 ℃, measured at a point not more than 1 m from the sample and at the same height as the sample. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Luminous flux, luminaire efficacy, zonal lumen were calculated from the software taken at 1 °vertical intervals and 22.5 °horizontal intervals.
2) Chromaticity Measurement – Sphere-Spectroradiometer Method: Chromaticity parameters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 ºC ±1 ºC. The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral power distribution taken at 5 nm intervals over the range of 380 to 780 nm.
3) Electrical Measurements: Electrical parameters were measured using power meters incorporated in goniophotometer or sphere-spectroradiometer system. The ambient temperature surrounding the sample was maintained at 25 ºC ±1 ºC. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Voltage, frequency, current, power, power factor and total harmonic distortion were measured by and read from the power meter.

2.1 Summary of Test Result

Criteria Item for 2.4m	Measured Value		Compliance	Requirement (DLC V5.1)	
Minimum Total Luminous	10146		Pass	>=375(lm/ft) (-10%)	
Minimum Luminous Efficacy	123.12		Pass	Standard: >= 115(-3%)	Premium: >= 130(-3%)
Minimum Power Factor	0.9473		Pass	>= 0.9(-3%)	
Maximum THD %	12.94		Pass	<= 20(+5)	
Minimum CRI*	84.90		Pass	>= 80(-1)	
Minimum R9*	20.1		Pass	>= 0(-1)	
Minimum Rg*	90.99		Pass	>= 89(-1)	
Minimum Rf*	82.63		Pass	>= 70(-1)	
Rcs, h1%*	-12		Pass	-12%-23%(-1%)	
CCT (K)*	3500K	3425	Pass	<=6500K	
	5000K	5059			
	6500K	6458			
Zonal Lumen Requirement	0-60°	62.9	Pass	>=40(-3)	

Note: The data with an "*" means that 2.4m of integrating sphere data is from Test Laboratory UL Verification Services (Guangzhou) Co., Ltd.

2.2 Electrical, Photometric and Chromaticity Measurements

Note: TS-ST4-4FT40W-H1-TCP and TS-ST4-8FT80W-H1-TCP have the same cross section structure.

Test date	2023-08-16	**Test Ambient:**	25±1 °C
Test Orientation	As intended	**Stabilization Time (min)**	60
Model Number	TS-ST4-8FT80W-H1-TCP(0%,3500K)	**Total Operating Time(min)**	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE230802-B1	120.0	60	0.6884	82.09	0.9938	6.12
	277.0	60	0.3141	82.41	0.9473	12.94

Photometric Measurement – Goniophotometer Method(Test Distance: 26.00m):

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	10210**	10146**
Luminous Efficacy (lm/W)	124.38	123.12
Zonal lumens in the 0-60 °	62.9	--
Beam Angle (°)	123.3	--
Center Beam Candle Power (cd)	2719	--

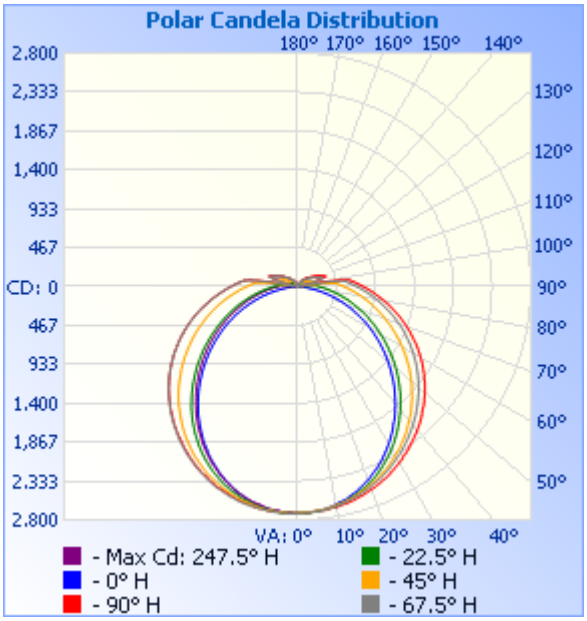
**The calculation method is on page 24.

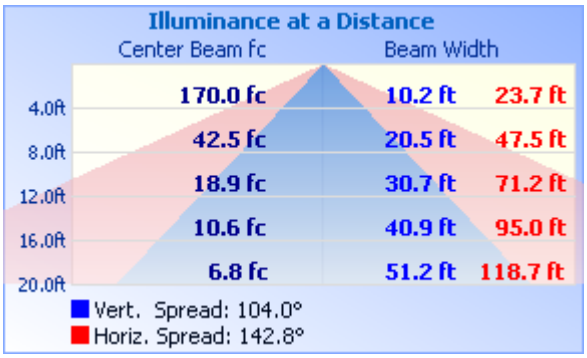
Zonal Lumen Tabulation

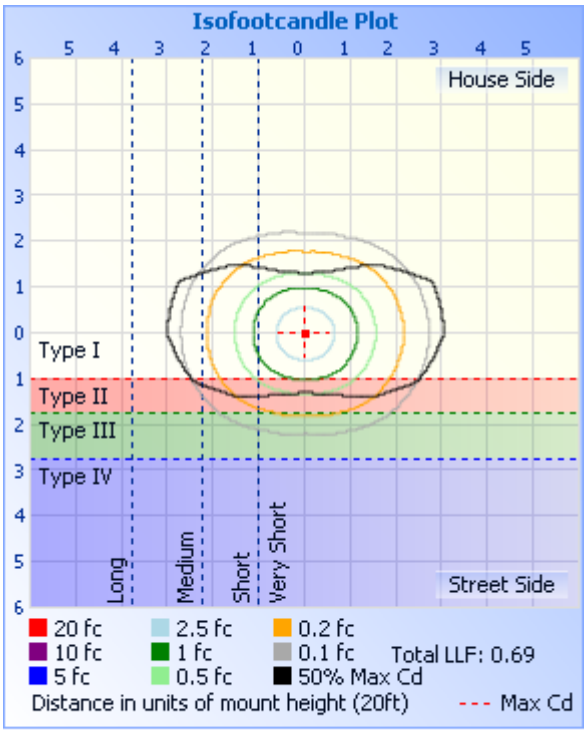
Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	2,130.5	20.9%
0-40	3,517.4	34.5%
0-60	6,423.5	62.9%
60-90	2,865.0	28.1%
70-100	2,037.9	20%
90-120	702.5	6.9%
0-90	9,288.5	91%
90-180	921.1	9%
0-180	10,209.6	100%

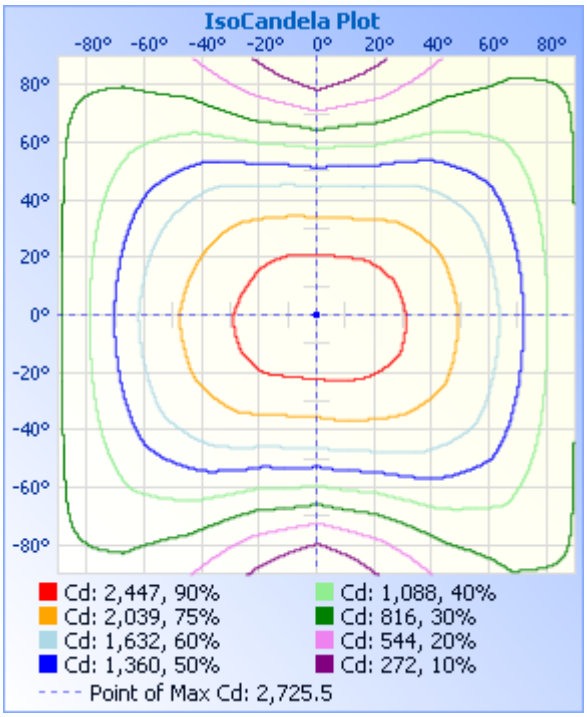
Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	%Total
0-10	257.5	2.5%	90-100	406.2	4%
10-20	740.7	7.3%	100-110	130.1	1.3%
20-30	1,132.3	11.1%	110-120	166.2	1.6%
30-40	1,386.9	13.6%	120-130	104.0	1%
40-50	1,482.8	14.5%	130-140	60.8	0.6%
50-60	1,423.3	13.9%	140-150	35.1	0.3%
60-70	1,233.4	12.1%	150-160	11.3	0.1%
70-80	959.2	9.4%	160-170	5.9	0.1%
80-90	672.5	6.6%	170-180	1.5	0%

Photometric Data









	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5	360
0	2719	2719	2719	2719	2719	2719	2719	2719	2719	2719	2719	2719	2719	2719	2719	2719	2719
1	2721	2718	2713	2721	2716	2721	2715	2716	2723	2720	2718	2724	2722	2722	2713	2716	2721
2	2718	2712	2709	2716	2716	2723	2721	2714	2717	2722	2718	2725	2722	2718	2711	2711	2718
3	2711	2713	2710	2712	2710	2719	2717	2714	2720	2721	2717	2721	2717	2717	2711	2711	2711
4	2705	2708	2700	2710	2707	2714	2713	2709	2714	2719	2713	2720	2716	2714	2708	2707	2705
5	2702	2706	2696	2702	2702	2709	2717	2708	2706	2713	2709	2720	2708	2719	2706	2696	2702
6	2698	2697	2692	2697	2697	2717	2712	2702	2706	2705	2708	2718	2713	2711	2700	2694	2698
7	2689	2696	2688	2694	2702	2704	2711	2691	2697	2705	2702	2711	2709	2704	2694	2688	2689
8	2677	2684	2682	2690	2696	2696	2697	2686	2685	2694	2704	2707	2709	2701	2692	2681	2677
9	2668	2678	2671	2678	2683	2695	2694	2674	2674	2687	2696	2703	2701	2698	2684	2666	2668
10	2652	2665	2661	2675	2685	2690	2687	2667	2666	2680	2685	2702	2694	2687	2671	2652	2652
11	2637	2653	2653	2664	2679	2677	2682	2656	2652	2673	2677	2691	2693	2689	2666	2645	2637
12	2626	2644	2646	2656	2666	2671	2669	2639	2637	2656	2670	2691	2682	2681	2657	2635	2626
13	2614	2632	2632	2643	2663	2668	2660	2627	2624	2642	2659	2684	2671	2671	2641	2618	2614
14	2591	2613	2623	2631	2652	2659	2651	2613	2610	2634	2653	2677	2669	2662	2633	2603	2591
15	2573	2599	2608	2622	2648	2647	2639	2592	2590	2626	2643	2671	2658	2651	2618	2588	2573
16	2556	2587	2596	2609	2631	2642	2626	2582	2572	2603	2634	2663	2651	2637	2606	2573	2556
17	2534	2565	2582	2597	2624	2628	2617	2562	2557	2587	2621	2650	2640	2628	2585	2552	2534
18	2513	2550	2569	2583	2616	2617	2598	2546	2531	2572	2608	2645	2627	2617	2582	2536	2513
19	2489	2533	2550	2570	2598	2606	2588	2523	2511	2554	2592	2630	2615	2605	2570	2514	2489
20	2467	2511	2536	2553	2588	2591	2571	2500	2488	2537	2583	2619	2601	2593	2550	2494	2467
21	2441	2489	2517	2538	2579	2577	2558	2480	2464	2518	2562	2604	2596	2580	2534	2470	2441
22	2414	2467	2505	2521	2563	2562	2541	2461	2439	2495	2550	2597	2592	2568	2514	2448	2414
23	2391	2444	2484	2509	2551	2550	2523	2430	2416	2471	2531	2582	2571	2553	2498	2423	2391
24	2365	2417	2467	2492	2537	2531	2509	2407	2386	2451	2518	2570	2559	2533	2486	2396	2365
25	2333	2390	2447	2473	2523	2517	2487	2383	2359	2429	2502	2554	2539	2519	2465	2374	2333
26	2307	2364	2428	2457	2510	2503	2468	2358	2331	2404	2485	2534	2522	2499	2441	2351	2307
27	2275	2335	2406	2438	2486	2485	2443	2328	2303	2378	2464	2522	2516	2483	2422	2321	2275
28	2245	2308	2384	2415	2472	2469	2426	2295	2275	2355	2444	2505	2496	2469	2400	2292	2245
29	2213	2281	2363	2399	2455	2450	2408	2271	2240	2326	2425	2489	2480	2448	2377	2263	2213
30	2181	2248	2340	2376	2437	2432	2383	2240	2211	2299	2403	2471	2461	2433	2357	2232	2181

31	2147	2221	2316	2359	2425	2407	2360	2207	2182	2269	2383	2454	2441	2409	2334	2204	2147
32	2112	2189	2298	2342	2398	2389	2336	2176	2146	2241	2361	2433	2424	2391	2313	2169	2112
33	2080	2155	2270	2314	2383	2372	2312	2145	2112	2206	2337	2417	2404	2372	2288	2137	2080
34	2043	2126	2243	2296	2364	2349	2290	2113	2076	2178	2312	2399	2383	2346	2264	2104	2043
35	2005	2088	2222	2270	2343	2329	2260	2081	2043	2145	2289	2373	2367	2326	2235	2072	2005
36	1974	2057	2193	2250	2319	2313	2232	2047	2008	2112	2262	2355	2342	2305	2210	2040	1974
37	1938	2022	2172	2225	2300	2285	2209	2012	1972	2077	2239	2333	2320	2280	2184	2003	1938
38	1900	1986	2140	2201	2279	2262	2178	1975	1936	2049	2213	2312	2299	2260	2155	1972	1900
39	1862	1953	2109	2180	2260	2236	2150	1943	1894	2013	2185	2289	2281	2239	2129	1937	1862
40	1824	1917	2084	2155	2236	2216	2120	1907	1856	1975	2158	2266	2258	2217	2097	1900	1824
41	1782	1880	2057	2129	2215	2190	2088	1872	1819	1943	2127	2243	2236	2190	2069	1864	1782
42	1743	1845	2032	2106	2192	2164	2063	1839	1781	1904	2100	2220	2213	2160	2038	1827	1743
43	1706	1809	1999	2081	2169	2141	2029	1800	1744	1868	2070	2197	2191	2138	2011	1790	1706
44	1668	1772	1967	2055	2144	2117	1996	1762	1706	1831	2043	2172	2161	2111	1979	1752	1668
45	1628	1731	1938	2029	2113	2087	1966	1724	1664	1794	2014	2148	2138	2085	1947	1715	1628
46	1588	1698	1911	2002	2093	2062	1935	1689	1623	1754	1985	2123	2118	2059	1919	1677	1588
47	1543	1657	1873	1975	2071	2036	1905	1649	1581	1718	1954	2098	2092	2033	1890	1639	1543
48	1507	1621	1843	1949	2044	2010	1870	1614	1544	1677	1923	2069	2063	2008	1857	1602	1507
49	1465	1581	1815	1923	2015	1982	1838	1574	1502	1639	1892	2045	2042	1978	1828	1566	1465
50	1422	1543	1783	1894	1993	1953	1802	1535	1458	1598	1860	2018	2015	1954	1790	1524	1422
51	1386	1504	1748	1866	1964	1926	1768	1498	1420	1564	1828	1989	1989	1924	1759	1486	1386
52	1341	1467	1716	1836	1938	1897	1735	1459	1378	1524	1795	1963	1963	1896	1726	1449	1341
53	1300	1427	1684	1808	1912	1867	1703	1420	1336	1483	1760	1933	1933	1868	1696	1408	1300
54	1258	1389	1655	1781	1879	1838	1667	1382	1293	1445	1730	1906	1905	1840	1662	1369	1258
55	1217	1349	1615	1753	1853	1808	1632	1343	1251	1406	1697	1878	1880	1810	1629	1331	1217
56	1176	1311	1586	1722	1824	1781	1597	1305	1209	1365	1664	1849	1852	1781	1593	1292	1176
57	1133	1272	1549	1694	1793	1750	1564	1266	1169	1326	1630	1820	1822	1751	1563	1254	1133
58	1093	1234	1520	1662	1767	1718	1528	1229	1126	1285	1598	1792	1791	1724	1529	1216	1093
59	1050	1196	1483	1631	1737	1688	1494	1191	1083	1247	1563	1759	1763	1691	1493	1178	1050
60	1010	1158	1451	1603	1703	1655	1458	1153	1042	1209	1530	1730	1733	1661	1462	1138	1010
61	970	1120	1417	1572	1676	1625	1424	1114	1000	1170	1497	1701	1703	1629	1426	1099	970
62	927	1082	1383	1541	1643	1594	1390	1076	958	1131	1462	1669	1673	1598	1393	1061	927

63	886	1045	1346	1511	1614	1562	1355	1039	919	1092	1431	1638	1639	1568	1360	1022	886
64	842	1009	1315	1479	1582	1529	1320	1003	875	1055	1396	1605	1608	1536	1328	986	842
65	801	971	1278	1447	1549	1496	1288	966	833	1018	1362	1574	1579	1505	1293	948	801
66	761	934	1246	1416	1519	1465	1255	927	791	980	1329	1542	1547	1474	1257	911	761
67	720	898	1215	1389	1488	1432	1221	891	749	944	1295	1509	1513	1441	1227	875	720
68	679	862	1179	1358	1455	1399	1187	855	709	908	1264	1479	1482	1408	1193	839	679
69	637	825	1146	1325	1424	1367	1154	819	668	872	1229	1446	1449	1379	1159	803	637
70	598	790	1114	1293	1391	1336	1123	783	628	836	1196	1415	1417	1344	1126	769	598
71	558	757	1084	1257	1359	1303	1090	748	587	800	1163	1382	1384	1312	1092	734	558
72	517	722	1051	1228	1327	1271	1058	715	547	766	1131	1350	1354	1280	1061	700	517
73	477	691	1020	1197	1293	1238	1025	682	506	732	1098	1318	1321	1248	1031	668	477
74	440	657	989	1166	1263	1208	994	648	468	697	1066	1285	1289	1214	998	634	440
75	402	626	959	1135	1231	1175	963	616	428	665	1034	1254	1254	1182	967	603	402
76	365	596	930	1102	1199	1144	932	586	391	634	1003	1220	1223	1152	937	572	365
77	329	565	899	1073	1167	1111	902	554	352	602	972	1188	1189	1120	904	542	329
78	293	536	870	1042	1136	1082	873	526	316	571	940	1156	1156	1088	876	512	293
79	259	508	841	1014	1104	1051	842	494	281	542	911	1124	1123	1057	846	484	259
80	226	481	813	984	1072	1020	813	468	247	513	881	1093	1093	1025	817	457	226
81	195	454	785	955	1043	989	785	441	214	487	852	1061	1062	994	790	431	195
82	164	428	758	925	1013	958	757	415	184	460	823	1030	1032	965	760	405	164
83	136	403	731	897	983	929	731	391	153	436	794	1000	1000	935	733	381	136
84	111	380	704	870	954	902	705	367	126	411	767	971	971	906	707	358	111
85	89	357	679	843	926	874	678	344	101	388	739	941	941	878	681	336	89
86	68	337	654	816	897	846	653	322	79	366	713	913	913	851	655	316	68
87	52	317	629	791	870	819	628	302	61	345	688	884	883	824	630	297	52
88	40	297	606	765	843	792	605	282	46	326	663	857	856	797	607	278	40
89	32	280	582	739	817	765	581	264	36	307	639	830	830	771	583	261	32
90	29	264	560	714	791	739	558	247	30	290	617	804	802	748	561	244	29
91	29	249	538	690	766	714	536	231	28	274	595	780	777	723	540	230	29
92	29	234	519	667	742	689	515	216	28	259	573	756	751	698	519	216	29
93	29	218	499	646	719	666	494	201	28	243	551	731	729	674	500	192	29
94	29	169	479	625	696	642	474	163	28	205	531	706	705	653	481	138	29

95	29	106	460	605	674	621	454	106	28	140	509	683	682	630	459	92	29
96	29	84	414	585	652	599	421	84	27	93	466	661	659	608	383	81	29
97	29	86	320	563	630	576	331	83	27	85	370	632	637	570	287	87	29
98	30	95	210	491	605	516	225	86	27	91	261	541	576	469	175	87	30
99	30	89	94	384	512	402	112	80	27	96	143	421	467	356	74	84	30
100	30	86	51	268	399	287	62	77	28	94	62	299	354	240	57	82	30
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102	30	83	84	28	163	40	92	73	28	90	88	49	121	17	117	78	30
103	30	81	137	5	42	5	143	72	28	89	136	9	16	8	171	77	30
104	30	79	200	6	5	7	199	70	29	87	192	12	5	28	219	75	30
105	30	77	208	59	6	58	205	67	29	86	240	73	17	95	217	73	30
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112	28	66	181	289	339	292	174	60	28	73	206	327	342	301	185	64	28
113	27	64	175	279	327	282	168	59	27	72	200	317	330	292	179	62	27
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117	22	55	155	242	283	244	145	52	25	64	173	278	285	255	154	54	22
118	21	52	149	234	272	234	140	49	24	62	166	268	274	246	148	52	21
119	19	50	144	225	262	226	135	47	22	60	160	259	264	237	143	49	19
120	17	48	139	217	252	218	130	45	21	58	154	250	255	228	138	47	17
121	16	46	134	209	243	210	125	43	19	56	149	242	246	221	134	45	16
122	14	44	130	202	234	202	121	41	18	53	144	233	236	213	130	44	14
123	13	42	125	194	225	194	117	38	17	52	139	224	228	205	127	42	13
124	12	40	121	187	217	186	113	37	15	49	135	216	220	197	123	41	12
125	12	39	117	180	209	179	110	35	14	47	131	208	212	190	120	40	12
126	11	38	113	173	201	172	106	33	13	45	127	200	204	183	116	39	11

127	10	37	109	167	193	166	102	30	12	44	123	192	196	176	113	38	10
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129	10	33	101	154	179	153	95	10	11	42	116	177	181	163	107	32	10
130	11	16	98	148	171	148	92	10	10	41	113	171	174	157	104	15	11
131	12	14	94	143	165	142	89	10	10	40	109	164	168	151	101	14	12
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146	24	33	41	76	88	78	51	25	21	32	59	88	89	83	60	34	24
147	24	31	6	72	83	74	47	25	21	30	17	83	85	79	33	31	24
148	23	30	6	67	79	71	40	24	21	30	11	79	82	75	9	30	23
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150	23	28	5	61	70	64	28	22	20	25	11	71	74	68	8	28	23
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159	17	17	26	6	10	4	30	20	14	17	20	39	3	15	30	16	17
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179	19	16	14	13	11	10	13	13	15	19	18	14	11	13	14	15	19
180	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19

2.3 Electrical, Photometric and Chromaticity Measurements

Test date	2023-08-16	**Test Ambient:**	25±1 °C
Test Orientation	As intended	**Stabilization Time (min)**	60
Model Number	TS-ST4-8FT80W-H1-TCP(0%,3500K)	**Total Operating Time(min)**	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE230802-B1	120.0	60	0.6916	82.49	0.9940	6.10
	277.0	60	0.3156	82.81	0.9474	12.93

Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.2637)(4π geometry):

Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	3425
Duv	-0.00082
Chromaticity (x, y)	x=0.4087 y=0.3905
Chromaticity (u', v')	u^=0.2380 v^=0.5117
Color Rendering Index (CRI)	84.9
R9	20.1
Rg	95.49
Rf	84.92
Rcs,h1	-11

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	10300	10247
Luminous Efficacy (lm/W)	124.86	123.74

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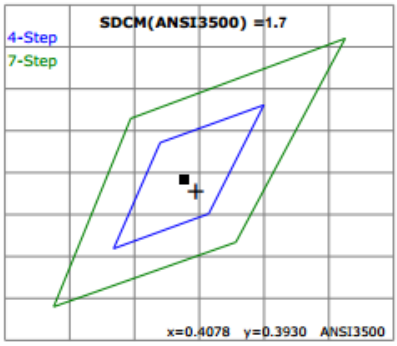
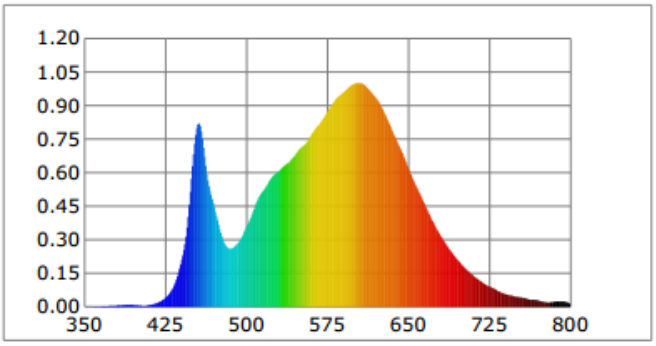
Spectral Power Distribution & Chromaticity Diagram





Color Rendering Index: Ra=84.9 Color Fidelity: Rf=84.92 Color Gamut: Rg=95.49 GAI:95.5

R1=84.0 R2=92.3 R3=96.5 R4=82.4 R5=83.4 R6=89.0 R7=85.6 R8=66.0
R9=20.1 R10=80.8 R11=81.2 R12=65.6 R13=86.2 R14=98.2 R15=74.8

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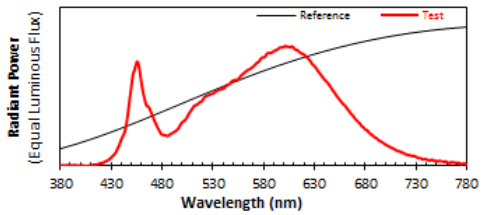
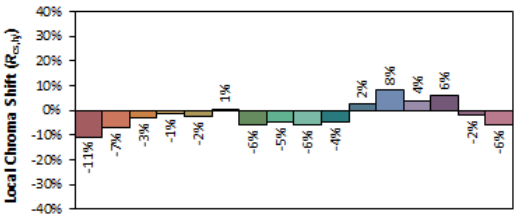
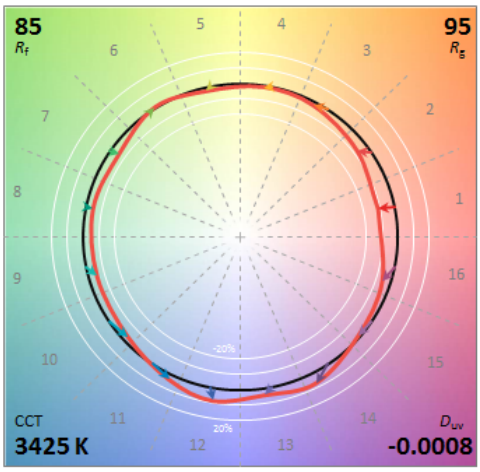
TM30

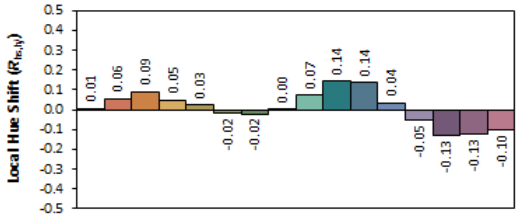
ANSI/IES TM-30-18 Color Rendition Report

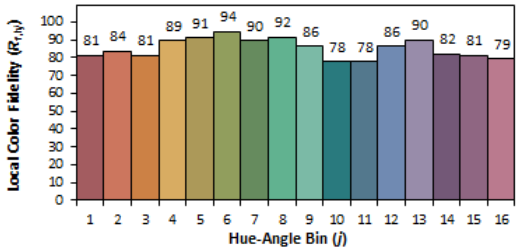
Source:	BXEN-35E-21M-3C	Manufacturer:	Torshare Ltd.
Date:	2023-08-16	Model:	TS-ST4-8FT80W-H1-TCP(0%,3500K)

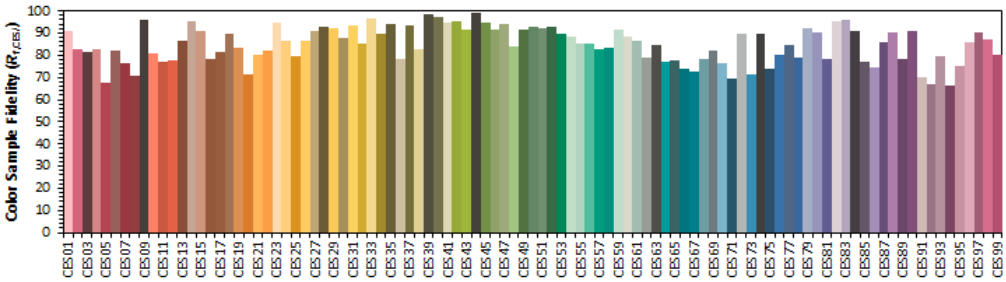












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.4087
y	0.3905
u'	0.2380
v'	0.5117

CIE 13.3-1995 (CRI)	
R_a	85
R_9	20

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

2.4 Electrical, Photometric and Chromaticity Measurements

Test date	2023-08-16	**Test Ambient:**	25±1 °C
Test Orientation	As intended	**Stabilization Time (min)**	60
Model Number	TS-ST4-8FT80W-H1-TCP(50%,5000K)	**Total Operating Time(min)**	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE230802-B1	120.0	60	0.6804	81.15	0.9939	6.11
	277.0	60	0.3104	81.47	0.9475	12.92

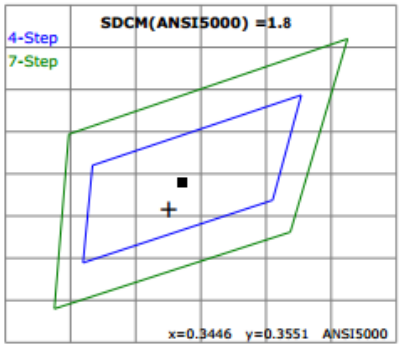
Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.2641)(4π geometry):

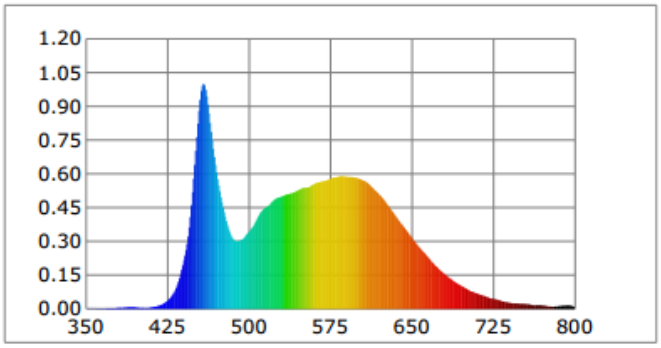
Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	5059
Duv	0.00034
Chromaticity (x, y)	x=0.3435 y=0.3510
Chromaticity (u', v')	u^=0.2106 v^=0.4841
Color Rendering Index (CRI)	86.77
R9	27.9
Rg	93.88
Rf	84.47
Rcs,h1	-10

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	10742	10674
Luminous Efficacy (lm/W)	132.37	131.02

Spectral Power Distribution & Chromaticity Diagram





Color Rendering Index: Ra=86.77 Color Fidelity: Rf=84.47 Color Gamut: Rg=93.88 GAI:90.98

R1=87.0 R2=95.2 R3=95.2 R4=83.1 R5=85.9 R6=89.9 R7=86.4 R8=71.5
R9=27.9 R10=86.4 R11=82.7 R12=63.9 R13=90.2 R14=98.4 R15=79.2

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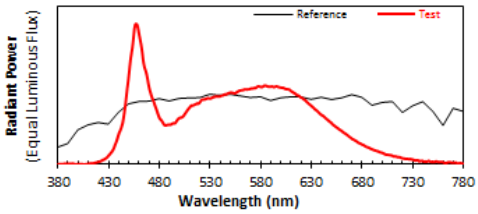
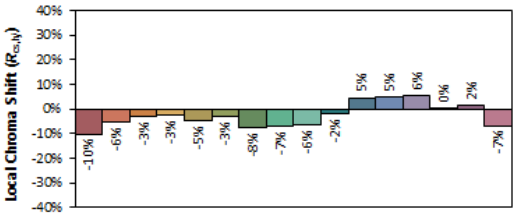
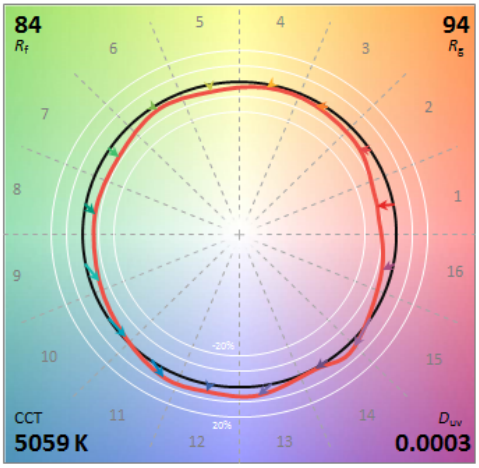
TM30

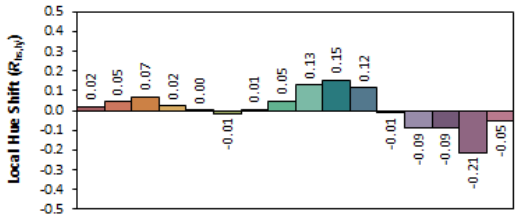
ANSI/IES TM-30-18 Color Rendition Report

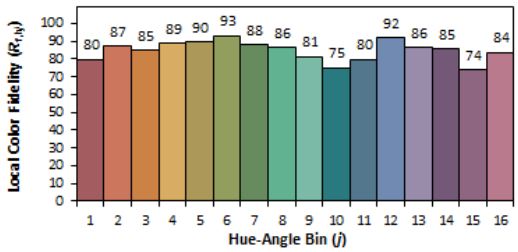
Source:	BXEN-35E-21M-3C BXEN-65E-21M-3C	Manufacturer:	Torshare Ltd.
Date:	2023-08-16	Model:	TS-ST4-8FT80W-H1-TCP(50%, 5000K)

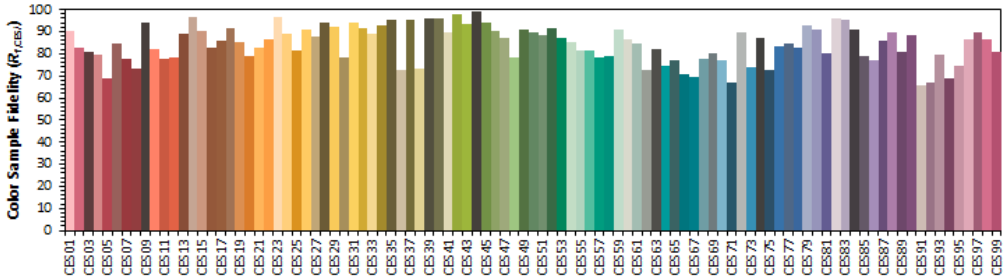












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3435
y	0.3510
u'	0.2106
v'	0.4841

CIE 13.3-1995 (CRI)

R_a	87
R_9	28

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

2.5 Electrical, Photometric and Chromaticity Measurements

Test date	2023-08-16	**Test Ambient:**	25±1 °C
Test Orientation	As intended	**Stabilization Time (min)**	60
Model Number	TS-ST4-8FT80W-H1-TCP(100%,6500K)	**Total Operating Time(min)**	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE230802-B1	120.0	60	0.6853	81.72	0.9937	6.13
	277.0	60	0.3126	82.04	0.9476	12.91

Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.2649)(4π geometry):

Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	6458
Duv	0.00476
Chromaticity (x, y)	x=0.3131 y=0.3324
Chromaticity (u', v')	u^=0.1968 v^=0.4702
Color Rendering Index (CRI)	85.5
R9	22.7
Rg	90.99
Rf	82.63
Rcs,h1	-12

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	10664	10597
Luminous Efficacy (lm/W)	130.49	129.17

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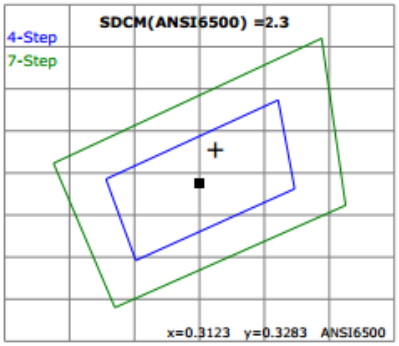
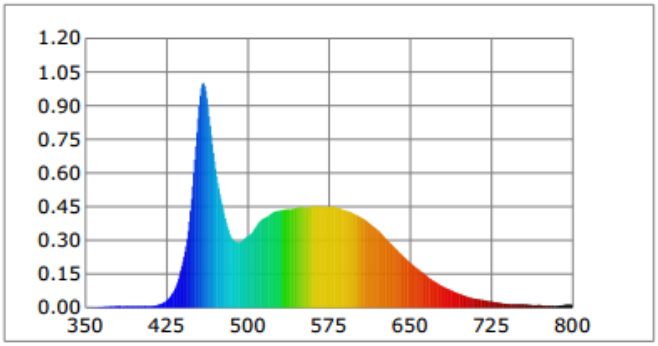
Spectral Power Distribution & Chromaticity Diagram





Color Rendering Index: Ra=85.5 Color Fidelity: Rf=82.63 Color Gamut: Rg=90.99 GAI:89.49

R1=85.6 R2=95.2 R3=94.3 R4=79.2 R5=83.6 R6=89.4 R7=85.8 R8=71.1
R9=22.7 R10=86.2 R11=79.6 R12=58.2 R13=89.6 R14=97.6 R15=78.1

Ningbo TengLi Testing Co., Ltd

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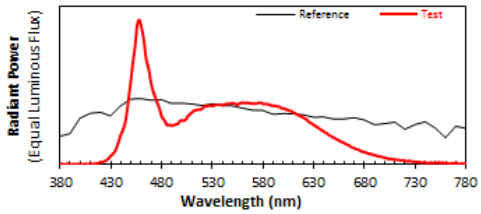
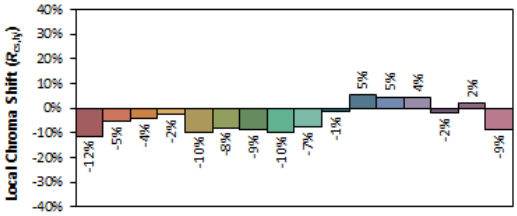
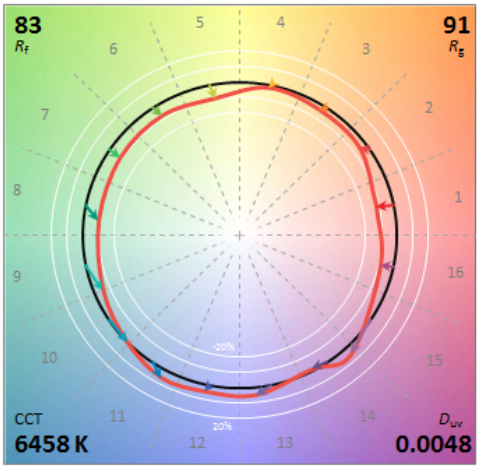
TM30

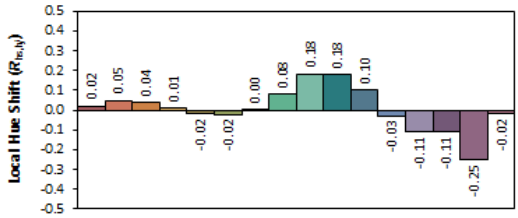
ANSI/IES TM-30-18 Color Rendition Report

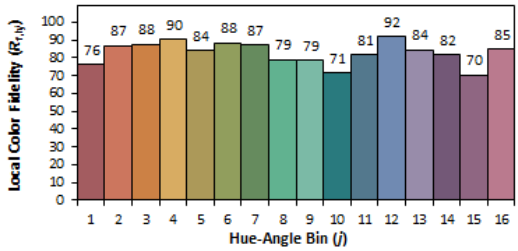
Source:	BXEN-65E-21M-3C	Manufacturer:	Torshare Ltd.
Date:	2023-08-16	Model:	TS-ST4-8FT80W-H1-TCP(100%,6500K)

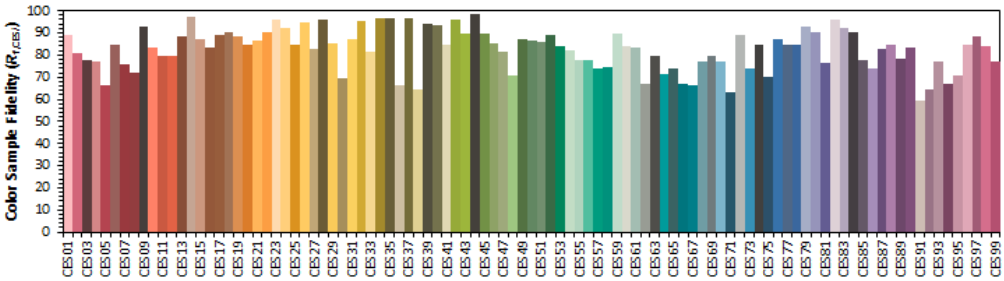












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3131
y	0.3324
u'	0.1968
v'	0.4702

CIE 13.3-1995 (CRI)

R_a	86
R_9	23

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

2.6 Liner product comparison for 1.2m and 2.4m

Model	TS-ST4-4FT40W-H1-TCP	TS-ST4-8FT80W-H1-TCP
Rated power	25W/30W/40W	50W/64W/80W
Size(m)	1.2m	2.4m
LED Spacing (mm)	2.5	2.5
LED QTY	504	1008
Driver Model Number	FS-TMG041B0880TWP-12V	FS-TMG039B1600TWP-12V
Calculation Method	Please refer to below “Calculation method”	
LED Working Current (mA)	47.8	47.9

Remark	TS-ST4-4FT40W-H1-TCP	TS-ST4-8FT80W-H1-TCP
Cross Section		
Overall Appearance		
Without lens		
lens		
The LED quantity,wattage and the length of TS-ST4-4FT40W-H1-TCP are half of TS-ST4-8FT80W-H1-TCP.		

Calculation method				
Model number	TS-ST4-4FT40W-H1-TCP		TS-ST4-8FT80W-H1-TCP	
Test Voltage (V)	120	277	120	277
Frequency (Hz)	60	60	60	60
Total Luminous of integrating sphere (lm)	5130	5098	10300	10247
Total Luminous of goniophotometer (lm)	5079.5	5047.6	10210**	10146**
Scale factor= 2.4m integrating sphere (lm)/ 1.2m integrating sphere (lm)	2.01	2.01	--	--

** This value is calculated and the calculation formula is as below:

10210=5079.5*2.01

10146=5047.6*2.01

3. Test Equipment

Equipment ID	Equipment Name	Last Calibration Date	Next Calibration Date
ST-R-702	2 meter Integrating Sphere	Verified by D204 standard lamp	
ST-R-701	Spectral analysis system HAAS-1200	Verified by D204 standard lamp	
ST-R-703	Standard Lamp D204	2022-12-29	2023-12-28
ST-R-704	Power Meter for Integrating Sphere	2022-12-29	2023-12-28
ST-R-707	Temperature Probe for Integrating Sphere	2023-01-03	2024-01-02
ST-R-714	Goniophotometer system	Verified by D908S standard lamp	
ST-R-710	Standard Lamp D908S	2022-12-29	2023-12-28
ST-R-711	Power Meter for Goniophotometer	2022-12-29	2023-12-28
ST-R-709	Hygrothermograph for Goniophotometer	2023-01-03	2024-01-02
Uncertainty(K=2): Photometric Measurement (Sphere):3.40% Chromaticity Measurement(Sphere):44.8K Photometric Measurement(Goniophotometer):3.64%			

4. Product Photo

******** END OF REPORT ********